23MCAP305	Mobile Application Development (Flutter)	L	T	P	J	S	C	Year of Introduction
		1		2		3	2	2023

Preamble: This course provides an introduction to mobile application development using the Flutter framework. Students will learn how to build cross-platform mobile applications for iOS and Android using a single codebase. The course covers topics such as Flutter architecture, widget-based UI development, state management, navigation and data storage. Through hands-on projects and assignments, students will gain practical experience in developing interactive and responsive mobile applications.

Prerequisite: Basic programming language

Course Outcomes: After the completion of the course the student will be able to

CO1	Understand the features and installation of Flutter
CO2	Solve basic programs using Dart constructs.
CO3	Develop simple mobile applications in Flutter using Dart Language.
CO4	Develop interactive Flutter App by using widgets, layout, gestures and animation
CO5	Develop mobile applications using database Connections

CO - PO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				3		2		3			
CO2	3				3		2		3			
CO3	3				3		2		3			
CO4	3				3		2		3			
CO5	3				3		2		3			

Assessment Pattern for Theory component

Bloom's Category	Continuous Assessment Tools			End Semester Examination
	Test1	Test2	Other tools	
Remember	✓	--	✓	✓
Understand	✓	--	✓	✓
Apply	✓	--	✓	✓
Analyse			✓	
Evaluate			✓	
Create			✓	

Assessment Pattern for Lab component

Bloom's Category	Continuous Assessment Tools	
	Class work	Test1
Remember		
Understand	✓	✓
Apply	✓	✓
Analyse	✓	✓
Evaluate	✓	
Create	✓	

Mark Distribution of CIA							
Course Structure [L-T-P-J]	Attendance	Theory [L- T]			Practical [P]		Total Marks
		Assignment	Test-1	Test-2	Class work	Lab Exam	
1-0-2-0	5	10	20	--	25	40	100

Total Marks distribution			
Total Marks	CIA (Marks)	ESE (Marks)	ESE Duration
100	100		

SYLLABUS

MODULE I : Introduction to Flutter (2hrs)

Features of Flutter, Advantages and Disadvantages of flutter, flutter architecture, Flutter Installation, Creating Simple Application in Android Studio.

MODULE II :Flutter Basics (2hrs)

Introduction to Dart Programming-Variables and Data types- Decision Making and Loops. Functions- Object Oriented Programming.

MODULE III :Developing Flutter UI(Widgets ,Layouts and Gestures) (3hrs)

Using Common Widgets: Safearea, Appbar, Column, Row, Container, Buttons, Text, Richtext, Form ,Images And Icon. Building Layouts: High Level View Of Layouts, Creating The Layout, Types of Layout Widgets. Applying Gestures: Setting Up Gesturedetector, Implementing The Draggable And Dragtarget Widgets

MODULE IV : Animation and Navigation on Flutter (3hrs)

Adding Animation To An App :Using Animated Container, Using Animated CrossFade, Using Animated Opacity. Creating An App's Navigation: Using the Navigator, Using the Named Navigator Route, Using the Bottom NavigationBar, Using the TabBar and TabBar View.

MODULE V : Creating Apps using Database (2 hrs)

Using Firebase with Flutter: Adding the Firebase Backend, Configuring the Firebase Project, Testing and Deploying of Flutter Application: Widget testing, Deploying Flutter Apps on Android / iOS.

Text books

1. Beginning Flutter a Hands-on Guide to App Development, Marco L. Napoli, Wiley, 2020.
2. Beginning App Development with Flutter: Create Cross-Platform Mobile Apps, By Rap Payne, 2019
3. Progressive Web Application Development by Example: Develop fast, reliable, and engaging user experiences for the web, Packt Publishing Limited ,2018
4. Building Progressive Web Apps,O'Reilly 2017

Reference books

1. Flutter in Action by Eric Windmill, MANING, 2019
2. Google Flutter Mobile Development Quick Start Guide.Packt,2019

Online References:

1. https://docs.flutter.dev/reference/tutorials
2. https://www.tutorialspoint.com/flutter/index.htm
3. https://www.w3schools.com/css/css_rwd_intro.asp

4. https://www.javatpoint.com/flutter		
COURSE CONTENTS AND LECTURE SCHEDULE		
No.		No. of Hours
MODULE 1: Introduction to Flutter (2hrs)		
1.1	Features of Flutter, Advantages and Disadvantages of flutter, flutter architecture	1
1.2	Flutter Installation, Creating Simple Application in Android Studio	1
MODULE II: Introduction to Dart Programming (2hrs)		
2.1	Introduction to Dart Programming-Variables and Data types	1
2.2	Functions- Object Oriented Programming	1
MODULE III: Common Widgets, Layouts, Gestures (3hrs)		
3.1	Using Common Widgets: SafeArea, Appbar, Column, Row, Container, Buttons, Text , Richtext, Form ,Images and Icon.	1
3.2	BUILDING LAYOUTS : high level view of layouts, Creating the layout , Types of layout widgets	1
3.3	APPLYING GESTURES: Setting Up GestureDetector, Implementing the Draggable and Dragtarget Widgets	1
MODULE IV: Apply Animation and Navigation (3hrs)		
4.1	Adding Animation To An App :Using Animated Container, Using Animated CrossFade, Opacity	1
4.2	Creating An App's Navigation: Using the Navigator, Using the Named Navigator Route	1
4.3	Creating An App's Navigation: Using the Bottom NavigationBar, Using the TabBar and TabBarView.	1
MODULE V:Database Connection (2hrs)		
5.1	Using Firebase with Flutter: Adding the Firebase Backend, Configuring	1
5.2	Deploying of Flutter Application: Widget testing, Deploying Flutter Apps on Android / iOS.	1

LESSON PLAN FOR LAB COMPONENT

No.	Topic	No. of Hours	Experiment
1	Installation(CO1)	2	a) To install and configure Flutter Environment b) To design Flutter UI by including common widgets.

2	Dart Programming language (CO2)	5	a) Program to print your name in Dart. b) Program in Dart that finds simple interest. c) Write a password generator in Dart. Be creative with how you generate passwords - strong passwords have a mix of lowercase letters, uppercase letters, numbers, and symbols. The passwords should be random, generating a new password every time the user asks for a new password. Include your run-time code in a main method. d) Randomly generate a 4-digit number. Ask the user to guess a 4-digit number. For every digit the user guessed correctly in the correct place, they have a "cow". For every digit the user guessed correctly in the wrong place is a "bull." Every time the user makes a guess, tell them how many "cows" and "bulls" they have. Once the user guesses the correct number, the game is over. Keep track of the number of guesses the user makes throughout the game and tell the user at the end.
3	Layout, Widgets ,gestures ,animation, navigation(CO3 and CO4) (Apply all these on your selected application)	8	a) To create an interactive Form using form widget b) To design a layout of Flutter App using layout widgets c) To include icons, images, charts in Flutter app d) To apply navigation, routing and gestures in Flutter App
4	Database	9	a) To Connect Flutter UI with fireBase database

			b) To test and deploy production ready Flutter App on Android platform

CO Assessment Questions	
1	Briefly explain flutter architecture Design and explain flutter UI using basic widgets
2	Write a program (using functions!) that asks the user for a long string containing multiple words. Print back to the user the same string, except with the words in backwards order.
3	Explain widgets and layout with suitable example.
4	Explain various animations and navigation methods available in flutter
5	List the detailed steps to connect your application with firebase